

From Foundations to Applications

2026

Architecture Objective Functions Data Construction
Training Evaluation Applications

2022

≤2021

2024

2025

2023

Architecture

CMCM

Hypergraph Transformer

SpeechT5

Flamingo

VLMo

Data Construction

Flickr30k Entities

Visual Genome

HowTo100M

Localized Narratives

RxR

ALIGN

Applications

FREEDOM

RT-2

SayPlan

VoxPoser

Mila MoleculeSTM

Architecture

OneLLM

AnyGPT

LLaMA-VID

LLaVA-UHD

4M-21

Dense Connector

Objective Functions

POVID

VideoCon

CLIP-DPO

mDPO

Data Construction

GLIP

LF-VILA

Applications

VLP-MABSA

MGCA

Architecture

FDT

BLIP-2

CLIPTrans

4M

LQAE

Uni-Code

MEAformer

Objective Functions

FormNetV2

SPECTRA

SRCL

SigLIP

UCoFiA

DPO

Data Construction

CoDet

DataComp

Training

EVA-CLIP

PaLI

InstructBLIP

LLaVA

Evaluation

ARO

VSR

POPE

SeeTRUE

SugarCrepe

Architecture

MoT

LLaVA-ST

UNIALIGN

M3-JEPA

TokenPacker

AlignVLM

Objective Functions

CHiP

GRAM

MIA-DPO

GOAL (Choi)

HA-DPO

OC-CLIP

Data Construction

HarmonySet

SPARCL

VideoComp

VisCon-100K

Align2LLaVA

OmniAlign-V

Training

CogCoM

Molmo

RLAIF-V

MM-RLHF

Parrot

LATTE

Applications

Qi et al.

LLM-CXR

CARZero

FETTLE

Applications

FigStep

Malenia

EarthDial

MMedPO

SafeVLA

ThinkAct

Architecture

GraphThinker

SpatialThinker

IGVA

SparseCut

mKG-RAG

Objective Functions

AdViP

PowerCLIP

Data Construction

IRIS

ARM-Thinker

ZINA

Training

AdPO

Molmo2

Evaluation

BLEND-Vis

VLMBias

Chart2Code

Evaluation

SIUO

SAIL

SVLTA

VidHalluc

MIRAGE

MME

Applications

LIP

ACoT-VLA

Counterfactual VLA

HaNoRec

MLLMRec

CrisiSense-RAG

Objective Functions

BLIP

InstructGPT

PyramidCLIP

Training

CoCa

Evaluation

VALSE

Winoground

Objective Functions

AVSA

Gat et al.

CLIP

TACo

TupleInfoNCE

ALBEF

Data Construction

Grounded Video Description

CoMMA

Landmark-RxR

Data Construction

ReferItGame

RefCOCOg

AVTS

MIL-NCE

Training

SeLaVi

XDC

MERLOT

Evaluation

BLEU

METEOR

CIDEr

SPICE

CHAIR

Applications

Action Genome

CoMIR

GLoRIA

VTASC

Data Construction

DFN

Kosmos-2

MetaCLIP

ULIP-2

VeCLIP

JEST

Data Construction

HarmonySet

SPARCL

VideoComp

VisCon-100K

Align2LLaVA

OmniAlign-V

Training

InternVL

RLHF-V

VLGuard

VLM-RLAIF

MM1

Cambrian-1

Applications

Qi et al.

LLM-CXR

CARZero

FETTLE

Applications

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ThinkAct